

# A two-valued obstruction to gluing Fréchet powers

A proposed solution of Massas's sheaf question

Complete proofs, local classification, and reproducible checks

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**Main result, in ZF.** On the category of all proper filters on  $\omega$  extending the cofinite filter, equipped with the dense topology,

$$F \mapsto A^\omega / \sim_F \quad \text{is a sheaf if and only if} \quad |A| \leq 1.$$

Two distinct values already suffice for failure. An explicit countably generated covering sieve carries compatible constant local sections with no global representative.

**Research status.** This is an unrefereed manuscript prepared with ChatGPT. It presents a complete mathematical argument, not a claim of independent expert certification. The motivating question is explicitly open in the cited 2023 article and 2024 dissertation. A targeted literature check found no later resolution, but priority and novelty have not been established. The computational supplements check exact finite consequences; they are not a formalization of the infinite proof.

## Abstract

Massas asks whether uncountability is necessary for the reduced-power presheaf associated with a structure to fail the sheaf condition on the Fréchet category. We give a negative answer using only two distinct elements. Partition  $\omega$  into infinitely many infinite rows. The filters concentrated on individual rows, together with the filter concentrated on tails of rows, generate a dense covering sieve. Assign one constant value on every row component and a different constant value on the tail component. These sections match because the components are incompatible. A global sequence would nevertheless have to take both values cofinitely on one infinite row. The construction and the resulting classification are proved in ZF. We also prove separatedness, a finite-pasting theorem showing that the countable number of generators is sharp, and failure above every countably generated free filter without any choice axiom. Under dependent choice, sheafness above an arbitrary base filter is equivalent, for a nontrivial value set, to finiteness of its quotient Boolean algebra. Finally, in ZFC, we identify the associated sheaf with a product of ultrapowers and explain the obstruction as a missing two-valued section. Every extension is proved independently of the literature, with its foundational assumptions stated explicitly.

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# 1 The question and the result

## 1.1 A specific published open problem

In his treatment of hyperreal numbers by possibility semantics, Massas considers a varying reduced power indexed by proper filters extending the cofinite filter on  $\omega$ . In [1, Definitions 5.4–5.7 and Fact 5.10], he proves that this presheaf need not be a sheaf. His counterexample assumes an injection  $2^\omega \hookrightarrow A$ ; with choice, he explains that the argument works for every uncountable domain. The question immediately after Fact 5.10 asks whether uncountability is necessary. It appears on printed page 517. The same question is repeated after Fact 7.4.10, on printed page 249 of his 2024 dissertation [2].

The issue concerns the *underlying set presheaf*. We write it as  $\mathcal{R}_A$  to distinguish it from a single ultrapower or a single ordered field. Its sections at a filter  $F$  are sequences in  $A$ , identified when they agree on an  $F$ -large set:

$$\mathcal{R}_A(F) = A^\omega / \sim_F, \quad f \sim_F g \iff \{j : f(j) = g(j)\} \in F. \quad (1)$$

A sheaf condition asks whether compatible sections given at enough stronger filters come from one section at the original filter.

**Theorem 1.1** (Global classification in ZF). *For every set  $A$ , the presheaf  $\mathcal{R}_A$  on the Fréchet category with its dense topology is a sheaf if and only if  $A$  has at most one element. If  $a_0, a_1 \in A$  are distinct, failure occurs at the cofinite filter on a covering sieve generated by countably many explicitly defined, countably generated filters. All local sections in the counterexample are represented by one of the two constant sequences  $\overline{a_0}$  and  $\overline{a_1}$ .*

The theorem answers the selected question negatively and supplies the exact cardinality boundary. For the usual convention that a first-order structure has a nonempty domain, the boundary is simply that the domain is a singleton. The proof is given in Section 3; no assertion about unpublished work or priority is needed for its validity.

## 1.2 Why the construction works

The key distinction is between being compatible as filters and imposing requirements on overlapping subsets of  $\omega$ . A filter on one row and a filter on tails of rows have no common proper extension. Consequently, a matching family is allowed to assign them different constant values. Nevertheless, any *single sequence* representing both sections would have to be constant with two different values on almost all of a suitable row. This is impossible.

The argument replaces the large family of different values used in the published cardinality obstruction by two values and a tail condition. The obstruction is therefore not a shortage of enough distinct values in the domain. It is a failure of countable patching for representatives of local equivalence classes.

## 1.3 Scope and foundations

Sections 2–7 work in classical ZF, with no axiom of choice. “Choice-free” here does not mean intuitionistic: ordinary classical case distinctions are used. Section 8 uses dependent choice only for the infinite direction of a local Boolean classification. Section 9 is explicitly in ZFC and uses ultrafilter extensions. Neither of those later assumptions enters the main theorem.

The article does not propose a new arithmetic of all hyperreal or surreal numbers, and does not address an ordinal-arithmetic conjecture. It solves a particular gluing question that arises in a construction of the hyperreal line. The mathematical claims below are proved in full; novelty remains a separate bibliographical issue.

## 2 The filter site and its reduced powers

### 2.1 Filters and positive sets

Write  $\omega = \{0, 1, 2, \dots\}$  and  $S^c = \omega \setminus S$ . A *proper filter* on  $\omega$  is a family  $F \subseteq \mathcal{P}(\omega)$  that contains  $\omega$ , omits  $\emptyset$ , is closed under finite intersections, and is upward closed under inclusion. The cofinite filter is

$$\text{Fr} = \{X \subseteq \omega : \omega \setminus X \text{ is finite}\}.$$

Throughout, a *free filter* means a proper filter containing  $\text{Fr}$ . Such a filter has no finite member: a finite member and its cofinite complement would have empty intersection.

Given a family of subsets with the finite-intersection property,  $\langle \mathcal{E} \rangle$  denotes the filter it generates. The same notation is used for an improper generated filter when discussing compatibility, but an improper filter is never an object of our site.

**Definition 2.1.** For a proper filter  $F$ , a set  $S \subseteq \omega$  is *F-positive* if  $S^c \notin F$ . Two free filters  $F, G$  are *compatible* if some proper free filter contains both; otherwise they are incompatible.

**Lemma 2.2** (Adjoining a set). *Let  $F$  be a proper filter and  $S \subseteq \omega$ . Then*

$$\langle F \cup \{S\} \rangle \text{ is proper} \iff S^c \notin F.$$

*When this generated filter is proper, its members are precisely the sets  $X$  for which  $C \cap S \subseteq X$  for some  $C \in F$ . If  $F$  is free,  $S$  is  $F$ -positive if and only if  $C \cap S$  is infinite for every  $C \in F$ .*

*Proof.* Finite intersections of generators have the form  $C$  or  $C \cap S$  with  $C \in F$ . The latter is empty for some  $C$  exactly when some member of  $F$  is contained in  $S^c$ , equivalently when  $S^c \in F$ . This proves the first assertion and the description of the generated filter.

Now suppose  $F$  is free and  $C \cap S$  is finite. Intersecting  $C$  with the cofinite complement of  $C \cap S$  produces an  $F$ -member disjoint from  $S$ , so  $S^c \in F$ . Conversely, if  $S^c \in F$ , its intersection with  $S$  is empty. These implications prove the final assertion.  $\square$

Two filters are incompatible exactly when there exist  $C \in F$  and  $D \in G$  with  $C \cap D = \emptyset$ . Indeed, such an intersection prevents a proper common extension; if no such intersection is empty,  $\langle F \cup G \rangle$  itself is a proper common extension.

### 2.2 Objects, arrows, and covering sieves

Let  $\mathcal{F}$  be the set of free filters on  $\omega$ . Regard it as a category with a unique arrow  $G \rightarrow F$  when  $G \supseteq F$ . Thus a stronger filter is the source of an arrow to a weaker one. For clarity we use actual set inclusion rather than a separate reversed-order symbol.

A *sieve at  $F$*  is a family  $D \subseteq \mathcal{F}$  such that every  $G \in D$  extends  $F$ , and every extension of a member of  $D$  is also in  $D$ . It is *covering* when

$$\text{for every } G \supseteq F \text{ in } \mathcal{F}, \text{ there is } H \supseteq G \text{ with } H \in D. \quad (2)$$

These are exactly the covers used in the question. The sieve formulation is also the standard language for Grothendieck topologies [3, Section 7.48].

For  $H \in \mathcal{F}$ , write

$$\uparrow H = \{G \in \mathcal{F} : H \subseteq G\}.$$

Here the arrow points upward in *inclusion*, not upward in the opposite categorical order. A sieve *generated* by filters  $H_i$  is  $\bigcup_i \uparrow H_i$ . A countably generated sieve need not be a countable set of filters. Separately, a filter is *countably generated* when it has a countable filter base. We will use both notions and keep them distinct.

**Lemma 2.3.** *The covering sieves just defined form a Grothendieck topology. Every covering sieve is nonempty.*

*Proof.* The maximal sieve of all extensions of  $F$  is dense. If  $D$  is a covering sieve at  $F$  and  $G \supseteq F$ , the sieve of members of  $D$  extending  $G$  is dense above  $G$ : apply (2) to an arbitrary extension of  $G$ .

For transitivity, suppose  $D$  covers  $F$  and  $E$  is a sieve at  $F$  such that  $E \cap \uparrow G$  covers  $G$  for each  $G \in D$ . Given  $H \supseteq F$ , choose an extension  $G \supseteq H$  in  $D$ , then an extension  $K \supseteq G$  in  $E$ . Thus  $E$  is dense. These are the maximality, pullback, and transitivity axioms. Applying density to  $F$  itself shows that a covering sieve cannot be empty. All choices in this argument are within a fixed finite chain of existential assertions.  $\square$

### 2.3 Sections and matching families

For sequences  $f, g : \omega \rightarrow A$ , put

$$E(f, g) = \{j \in \omega : f(j) = g(j)\}.$$

The relation in (1) is an equivalence relation. If  $G \supseteq F$ , equality modulo  $F$  implies equality modulo  $G$ ; hence there is a well-defined restriction map

$$\rho_{F,G} : \mathcal{R}_A(F) \longrightarrow \mathcal{R}_A(G), \quad \rho_{F,G}([f]_F) = [f]_G.$$

Identity and composition hold directly, so  $\mathcal{R}_A$  is a presheaf.

A *matching family* on a sieve  $D$  is a collection  $s_G \in \mathcal{R}_A(G)$ , one for each  $G \in D$ , such that

$$\rho_{G,H}(s_G) = s_H \quad \text{whenever } G, H \in D \text{ and } H \supseteq G.$$

An *amalgam* at  $F$  is  $s \in \mathcal{R}_A(F)$  with  $\rho_{F,G}(s) = s_G$  for every  $G \in D$ . The presheaf is a *sheaf* if every matching family on every covering sieve has exactly one amalgam. It is *separated* if it has at most one.

There is an equivalent way to prescribe data on generators. Given filters  $H_i$ , sections at them determine a matching family on the generated sieve provided they agree at every common proper extension. If a filter extends several generators, this agreement makes the definition independent of which generator is used. No agreement condition is imposed between incompatible generators.

### 3 The explicit two-valued obstruction

#### 3.1 Rows and their tail filter

Fix an explicit partition

$$\omega = \bigsqcup_{n < \omega} B_n \quad (3)$$

into infinite sets, and let

$$T_m = \bigcup_{n \geq m} B_n = \omega \setminus \bigcup_{n < m} B_n. \quad (4)$$

An arithmetic formula for the partition is supplied in Section 4; therefore no choice is hidden in its existence. Define the row filters and the tail filter by

$$F_n = \{X \subseteq \omega : B_n \setminus X \text{ is finite}\}, \quad (5)$$

$$F_\infty = \{X \subseteq \omega : \text{for some } m, T_m \setminus X \text{ is finite}\}. \quad (6)$$

Equivalently,

$$F_n = \langle \text{Fr} \cup \{B_n\} \rangle, \quad F_\infty = \langle \text{Fr} \cup \{T_m : m < \omega\} \rangle.$$

**Lemma 3.1.** *All these filters are proper and free. They are countably generated. The filters  $F_n$  are pairwise incompatible, and every  $F_n$  is incompatible with  $F_\infty$ .*

*Proof.* The defining families are upward closed and contain every cofinite set. Closure under intersections for  $F_\infty$  follows by taking the larger of two tail indices and the union of the two finite exceptional sets. No row or tail is finite, so neither definition includes the empty set. A countable base for  $F_n$  is  $\{B_n \setminus \{0, \dots, r-1\} : r < \omega\}$ ; a countable base for  $F_\infty$  uses  $T_m \setminus \{0, \dots, r-1\}$  for  $(m, r) \in \omega^2$ .

For  $n \neq k$ , the disjoint sets  $B_n \in F_n$  and  $B_k \in F_k$  preclude a proper common extension. Also  $B_n \in F_n$  and  $T_{n+1} \in F_\infty$  are disjoint. This proves every claimed incompatibility.  $\square$

#### 3.2 A countably generated dense sieve

Consider

$$D = \uparrow F_\infty \cup \bigcup_{n < \omega} \uparrow F_n. \quad (7)$$

It is a sieve at Fr, and its displayed components are disjoint.

**Lemma 3.2** (Density). *The sieve  $D$  covers Fr.*

*Proof.* Let  $G \supseteq \text{Fr}$  be any proper free filter. If some row  $B_n$  is  $G$ -positive, Lemma 2.2 says that

$$H = \langle G \cup \{B_n\} \rangle$$

is a proper free filter. It extends both  $G$  and  $F_n$ , so  $H \in D$ .

Otherwise  $B_n^c \in G$  for every  $n$ . Since  $G$  is closed under *finite* intersections, for each  $m$  it contains

$$\bigcap_{n < m} B_n^c = T_m.$$

It consequently contains  $F_\infty$ , and  $G$  itself belongs to  $D$ . Both cases provide the extension required by (2).  $\square$

The second case does not use countable completeness of  $G$ . For a fixed  $m$ , only  $m$  sets are intersected. Nor does the first case require a simultaneous choice of positive rows for all filters: the proof establishes an extension for one arbitrary  $G$  at a time.

### 3.3 A matching family with no amalgam

Fix distinct  $a_0, a_1 \in A$ , and let  $\overline{a_i}$  be the constant sequence with value  $a_i$ . For each  $G \in D$ , define

$$s_G = \begin{cases} [\overline{a_0}]_G, & \text{if } G \supseteq F_n \text{ for some } n, \\ [\overline{a_1}]_G, & \text{if } G \supseteq F_\infty. \end{cases} \quad (8)$$

Lemma 3.1 makes the cases disjoint. Every extension of a filter in a component remains in that component, so restriction preserves its assigned constant section. Thus (8) is a matching family.

**Proposition 3.3** (The contradiction). *The matching family (8) has no amalgam in  $\mathcal{R}_A(\text{Fr})$ .*

*Proof.* Suppose  $[f]_{\text{Fr}}$  were an amalgam. Its restriction to the tail filter would satisfy

$$E_1 := \{j : f(j) = a_1\} \in F_\infty.$$

By (6), there is  $m$  such that  $T_m \setminus E_1$  is finite. Its restriction to the row filter  $F_m$  would satisfy

$$E_0 := \{j : f(j) = a_0\} \in F_m,$$

so  $B_m \setminus E_0$  is finite. Because  $B_m \subseteq T_m$ , the set  $B_m \setminus E_1$  is also finite. The infinite set  $B_m$  therefore has an element in  $E_0 \cap E_1$ . At that element,  $f$  has both values  $a_0$  and  $a_1$ , contrary to their being distinct.  $\square$

*Proof of Theorem 1.1.* If  $A$  has two distinct elements, the preceding construction proves failure of sheafness. If  $A$  is a singleton,  $\mathcal{R}_A(F)$  is a singleton for every object and every matching family has one amalgam. If  $A = \emptyset$ , every  $\mathcal{R}_A(F)$  is empty because  $\omega$  is nonempty. Every covering sieve is nonempty, so it has no matching family with values in this presheaf; the sheaf condition is then vacuously true. These exhaust the cases in classical ZF.  $\square$

### 3.4 Why several tempting objections do not apply

The family is not assigning different values on two compatible pieces. The row and tail filters are incompatible because the tail filter contains *all* row tails, including the one disjoint from the chosen row. A putative representative at the tail filter, however, agrees with its constant value outside finitely many points of *one* tail. Selecting the index of that tail identifies a row on which the global contradiction occurs.

The proof does not exchange two universal quantifiers, assume a uniform exceptional set for all rows, or require that  $f$  be two-valued everywhere. It uses the tail index supplied by  $f$  and the exceptional set supplied for that single row. Other values of  $f$ , when  $A$  is larger, cannot remove the contradiction.

Finally,  $D$  is the full extension-closed family in (7), not just the listed generators. This distinction is essential for matching the covering definition in the original question.

## 4 An arithmetic realization and an exact witness

### 4.1 A partition determined by powers of two

Take

$$B_n = \{2^n(2k+1) - 1 : k < \omega\}. \quad (9)$$

Every positive integer has a unique factorization as a power of two times an odd integer. Applied to  $j + 1$ , this shows that (9) is a partition of  $\omega$  into infinite sets. In this realization,

$$T_m = \{j < \omega : 2^m \text{ divides } j + 1\}. \quad (10)$$

The first few sets are

$$\begin{aligned} B_0 &= \{0, 2, 4, 6, \dots\}, & B_1 &= \{1, 5, 9, 13, \dots\}, \\ B_2 &= \{3, 11, 19, 27, \dots\}, & B_3 &= \{7, 23, 39, 55, \dots\}. \end{aligned}$$

One may view the rows as the finite 2-adic valuations of  $j + 1$ ; no 2-adic analysis is needed.

## 4.2 Turning cofinite bounds into a contradictory index

Suppose the hypothetical amalgam provides integers  $m, N_0, N_1$  such that

$$\begin{aligned} j \in B_m, j \geq N_0 &\implies f(j) = a_0, \\ j \in T_m, j \geq N_1 &\implies f(j) = a_1. \end{aligned}$$

Let  $M = \max(N_0, N_1)$ ,  $r = 2^m - 1$ , and  $p = 2^{m+1}$ . Set

$$q = \max\left(0, \left\lceil \frac{M - r}{p} \right\rceil\right), \quad j = r + pq. \quad (11)$$

Then  $j \geq M$  and  $j = 2^m(2q + 1) - 1 \in B_m \subseteq T_m$ . Thus both displayed implications apply to this particular integer. In fact  $j$  is the least element of  $B_m$  above both cutoffs.

For example, with  $m = 3$ ,  $N_0 = 100$ , and  $N_1 = 150$ , formula (11) gives  $r = 7$ ,  $p = 16$ ,  $q = 9$ , and  $j = 151$ . The included program `obstruction_witness.py` computes these certificates using arbitrary-precision integers:

```
python3 obstruction_witness.py 3 100 150
```

The program cannot infer cofinite bounds from an arbitrary black-box infinite sequence. It verifies the final explicit implication once such bounds are supplied. The proof of their existence under the amalgamation hypothesis is mathematical, not computational.

## 5 Separatedness and finite patching

### 5.1 Local equality always determines global equality

**Theorem 5.1** (Separatedness, ZF). *For every set  $A$ ,  $\mathcal{R}_A$  is separated for the dense topology. More explicitly, if  $D$  covers  $F$  and  $[f]_G = [g]_G$  for every  $G \in D$ , then  $[f]_F = [g]_F$ .*

*Proof.* Put  $E = E(f, g)$ . If  $E \notin F$ , the set  $E^c$  is  $F$ -positive, and  $H = \langle F \cup \{E^c\} \rangle$  is proper. By density there is  $K \supseteq H$  in  $D$ . Local equality gives  $E \in K$ , while  $E^c \in K$  follows from  $H \subseteq K$ . Their empty intersection contradicts properness. Hence  $E \in F$ .  $\square$

This is the equality instance of the local-character phenomenon also present in the source construction [1, Lemma 5.8]. The new obstruction concerns existence of a global section, not failure of this uniqueness property.

## 5.2 Every finite compatible system has a representative

The next theorem is slightly more general than needed: density is not required for existence of a simultaneous representative.

**Theorem 5.2** (Finite pasting, ZF). *Let  $A$  be nonempty, and let  $F_1, \dots, F_r$  be proper filters on an index set  $I$ , where  $1 \leq r < \omega$ . Let  $s_i \in A^I / \sim_{F_i}$ . Suppose the restrictions of  $s_i$  and  $s_j$  agree at every proper common extension of  $F_i$  and  $F_j$ . Then some  $f : I \rightarrow A$  represents  $s_i$  modulo  $F_i$  for all  $i$ .*

*Proof.* Choose representatives  $f_i$  for the finitely many sections. For each pair  $i < j$  we claim there are  $C_{ij} \in F_i$  and  $C_{ji} \in F_j$  such that

$$C_{ij} \cap C_{ji} \subseteq E(f_i, f_j). \quad (12)$$

If  $F_i, F_j$  are incompatible, choose their members with empty intersection. If they are compatible, their generated common filter  $H_{ij} = \langle F_i \cup F_j \rangle$  is proper. Compatibility of the sections implies  $E(f_i, f_j) \in H_{ij}$ . By the description of a generated filter, it contains an intersection of one member of  $F_i$  and one member of  $F_j$ . This gives (12). All the selections just made are finite and are available in ZF.

For each  $i$ , intersect the finitely many  $C_{ij}$  on its side and call the result  $C_i \in F_i$ . For  $r = 1$ , take  $C_1 = I$ . Then

$$C_i \cap C_j \subseteq E(f_i, f_j) \quad (i \neq j).$$

Fix  $a \in A$  and define  $f(t)$  as follows. If  $t$  belongs to at least one  $C_i$ , use  $f_i(t)$  for the least such  $i$ ; otherwise use  $a$ . Whenever  $t \in C_i$ , the chosen value agrees with  $f_i(t)$  by the intersection property. Thus  $C_i \subseteq E(f, f_i)$ , so  $[f]_{F_i} = s_i$  for every  $i$ .  $\square$

**Corollary 5.3** (Finite covers and sharpness). *Every matching family for  $\mathcal{R}_A$  on a finitely generated covering sieve has a unique amalgam. For  $|A| \geq 2$ , there is a failure of amalgamation on a countably generated covering sieve. Hence a witness cannot have finitely many generators, while countably many suffice.*

*Proof.* Apply Theorem 5.2 to sections at the finite generating filters. Their restrictions determine the given matching family on the entire sieve. The resulting representative yields a section at the base filter, and Theorem 5.1 gives uniqueness. If  $A$  is empty there is no matching family on a nonempty cover. The second assertion is the construction in Section 3.  $\square$

The word “countable” in this corollary refers to the number of *sieve generators*, not the number of filters in the full covering sieve. In particular, the obstruction does not contradict finite pasting: any finite part of its prescribed data can be represented, but the whole cover cannot.

## 6 The obstruction above an arbitrary base filter

The partition proof extends whenever a base filter admits infinitely many disjoint positive sets. This is the local form that will support the later classification.

**Theorem 6.1** (Positive-row obstruction, ZF). *Let  $F \in \mathcal{F}$ . Suppose  $(B_n)_{n < \omega}$  is a sequence of pairwise disjoint  $F$ -positive subsets of  $\omega$ . They need not cover  $\omega$ . For every set  $A$  with two distinct elements,  $\mathcal{R}_A$  fails the sheaf condition at  $F$  on the site of all free filters extending  $F$ .*

*Proof.* Define

$$T_m = \omega \setminus \bigcup_{n < m} B_n, \quad H_n = \langle F \cup \{B_n\} \rangle, \quad H_\infty = \langle F \cup \{T_m : m < \omega\} \rangle.$$

Each  $H_n$  is proper by positivity. For the tail filter, any finite intersection of its added generators is some  $T_m$ . If  $C \in F$ , then  $C \cap T_m$  contains  $C \cap B_m$ , which is infinite. Thus  $H_\infty$  is proper as well. Notice its precise membership test:

$$X \in H_\infty \iff \text{some } C \in F \text{ and } m < \omega \text{ satisfy } C \cap T_m \subseteq X. \quad (13)$$

The row filters are pairwise incompatible, and  $H_n$  is incompatible with  $H_\infty$  because  $B_n \cap T_{n+1} = \emptyset$ .

The sieve

$$D_F = \uparrow H_\infty \cup \bigcup_n \uparrow H_n$$

is dense above  $F$ . Given  $G \supseteq F$ , a  $G$ -positive row may be adjoined properly. If none is positive, all row complements lie in  $G$ ; therefore all  $T_m$  lie in  $G$ , and  $G \supseteq H_\infty$ .

Assign  $[\overline{a_0}]$  on the row components and  $[\overline{a_1}]$  on the tail component. This is again a matching family. If  $[f]_F$  glues it, let  $E_i = \{j : f(j) = a_i\}$ . Membership  $E_1 \in H_\infty$  gives  $C \in F$  and  $m$  with  $C \cap T_m \subseteq E_1$ . Membership  $E_0 \in H_m$  gives  $C' \in F$  with  $C' \cap B_m \subseteq E_0$ . Since  $B_m$  is  $F$ -positive,

$$(C \cap C') \cap B_m$$

is infinite and hence nonempty. It is contained in both  $E_0$  and  $E_1$ , a contradiction.  $\square$

*Remark 6.2.* Using  $T_m = \bigcup_{n \geq m} B_n$  in this more general proof would be unjustified when the rows do not cover  $\omega$ : avoiding every row would not force membership of that union. The complement-of-a-finite-union definition is the correct one and includes any leftover points.

## 7 Every countably generated free base already fails

The main theorem uses a particularly simple base. In fact, no countably generated free base can repair the gluing problem.

**Lemma 7.1** (Disjoint positive rows from a countable base, ZF). *Every countably generated proper free filter on  $\omega$  has a sequence of pairwise disjoint positive sets.*

*Proof.* Fix a countable base and replace it by successive finite intersections to obtain a decreasing base

$$C_0 \supseteq C_1 \supseteq C_2 \supseteq \cdots.$$

Every  $C_s$  is infinite. At stage  $s$ , for  $n = 0, \dots, s$ , let  $b_{n,s}$  be the least element of  $C_s$  not used previously, including earlier choices within that stage. Only finitely many numbers have been used at this point, so the least available element exists. This is a recursion with a uniquely determined value at every step, not an invocation of dependent choice.

Let  $B_n = \{b_{n,s} : s \geq n\}$ . No chosen integer is ever reused, so these sets are pairwise disjoint. If  $X \in F$ , some  $C_m$  is contained in  $X$ . For every  $s \geq \max(m, n)$ ,

$$b_{n,s} \in C_s \subseteq C_m \subseteq X.$$

These are infinitely many distinct elements of  $B_n \cap X$ . Lemma 2.2 proves that  $B_n$  is  $F$ -positive.  $\square$

**Corollary 7.2 (ZF).** *If  $F$  is a countably generated proper free filter and  $|A| \geq 2$ , then  $\mathcal{R}_A$  fails sheafness at  $F$  on a countably generated covering sieve. Its generating filters can also all be taken countably generated.*

*Proof.* Apply Theorem 6.1 to the preceding lemma. Adding one set, or a countable decreasing family of sets, to a countable filter base leaves a countable base, provided the generated filter is proper. Properness was proved in Theorem 6.1.  $\square$

## 7.1 Restricting the site does not remove this obstruction

Let  $\mathcal{F}_{\text{cg}}$  be the category consisting only of countably generated proper free filters, with the same inclusion arrows and the relative dense topology. For an arbitrary  $G$  in this smaller category, adjoining a positive row still gives a countably generated proper extension. The other branch of the density argument uses  $G$  itself. Thus the sieve of extensions of the displayed generators *within*  $\mathcal{F}_{\text{cg}}$  is still covering, and the same family still fails to glue. This applies above every object of that category.

Consequently, the main phenomenon is not caused by admitting complicated uncountably generated filters or by the presence of ultrafilters. The elementary countable-base part of the site already exhibits it. This observation does not assert that every free filter is countably generated; no such assertion is used.

## 8 A local Boolean-algebra classification

### 8.1 The quotient attached to a base

Fix  $F \in \mathcal{F}$ . Define its dual ideal and quotient Boolean algebra by

$$\mathcal{I}_F = \{S \subseteq \omega : S^c \in F\}, \quad \mathbb{B}_F = \mathcal{P}(\omega) / \mathcal{I}_F. \quad (14)$$

Let  $q_F : \mathcal{P}(\omega) \rightarrow \mathbb{B}_F$  be the quotient map. The ideal is proper and contains every finite set. We have

$$q_F(S) = 0 \iff S^c \in F, \quad q_F(S) = 1 \iff S \in F. \quad (15)$$

Thus  $F$ -positive sets represent exactly the nonzero Boolean elements.

**Lemma 8.1** (Filters above  $F$ ). *Taking image and inverse image under  $q_F$  gives a bijection between proper filters on  $\omega$  extending  $F$  and proper filters of  $\mathbb{B}_F$ .*

*Proof.* If  $G \supseteq F$  and  $S \Delta T \in \mathcal{I}_F$ , then  $(S \Delta T)^c \in F \subseteq G$ . Whenever  $S \in G$ , intersecting  $S$  with this complement gives a member of  $G$  contained in  $T$ , so  $T \in G$ . Hence membership in  $G$  depends only on the quotient class. Its image is therefore a proper Boolean filter. Conversely, the inverse image of a proper Boolean filter is a proper filter on  $\omega$  and contains  $q_F^{-1}(1) = F$ . These operations are inverse by the same membership test.  $\square$

### 8.2 The finite case needs no choice

**Proposition 8.2** (Finite quotient implies sheafness, ZF). *If  $\mathbb{B}_F$  is finite, then for every set  $A$ , the restriction of  $\mathcal{R}_A$  to the category of filters extending  $F$  is a sheaf.*

*Proof.* A finite Boolean algebra has only finitely many proper filters. Lemma 8.1 shows that the category above  $F$  has finitely many objects. Thus any covering sieve at any of its objects is generated by finitely many filters, namely its own members. Corollary 5.3 proves the sheaf assertion. Empty value sets are covered by the same vacuity argument as in Theorem 1.1.  $\square$

There is a useful concrete description of this case. If  $\mathbb{B}_F$  has atoms  $b_1, \dots, b_k$ , let

$$U_i = \{X \subseteq \omega : b_i \leq q_F(X)\}.$$

Each  $U_i$  is an ultrafilter extending  $F$ : an atom lies below exactly one of a Boolean element and its complement. These are all the ultrafilter extensions of  $F$ , and

$$F = \bigcap_{i=1}^k U_i. \quad (16)$$

No ultrafilter lemma is needed to construct these finitely many filters.

Every proper filter of a finite Boolean algebra is generated by its least nonzero member. Therefore every  $G \supseteq F$  has the form  $G = \bigcap_{i \in J} U_i$  for a nonempty subset  $J \subseteq \{1, \dots, k\}$ . For nonempty  $A$  the natural map is a bijection:

$$A^\omega / G \longrightarrow \prod_{i \in J} A^\omega / U_i, \quad [f]_G \longmapsto ([f]_{U_i})_{i \in J}. \quad (17)$$

Injectivity follows from the expression for  $G$ . For surjectivity, take representatives for the finitely many given classes and paste them on disjoint representatives of the Boolean atoms. Such disjoint representatives exist by finite disjointification; the negligible leftover set can be assigned to the first piece. The resulting sequence has the prescribed class at each relevant ultrafilter. This is another instance of finite pasting. For  $A = \emptyset$  both sides are empty since  $J$  is nonempty.

### 8.3 The infinite case under dependent choice

**Lemma 8.3** (ZF + DC). *Every infinite Boolean algebra has a countably infinite sequence of pairwise disjoint nonzero elements.*

*Proof.* For an element  $r$ , write  $\mathbb{B} \restriction r$  for the interval of elements below  $r$ . Start with  $r_0 = 1$ , whose interval is infinite. An element  $r$  with infinite interval is not an atom, so it can be split into two disjoint nonzero elements  $c, d$  with  $c \vee d = r$ . At least one of the intervals below  $c, d$  is infinite: otherwise the map  $x \mapsto (x \wedge c, x \wedge d)$  would embed the interval below  $r$  into a finite product of finite sets.

Use dependent choice to repeat this operation, retaining an infinite interval as  $r_{n+1}$  and calling the other nonzero piece  $b_n$ . Then  $b_n \wedge r_{n+1} = 0$  and all later  $b_j$  lie below  $r_{n+1}$ . Hence the  $b_n$  are nonzero and pairwise disjoint.  $\square$

**Lemma 8.4** (ZF + DC). *If  $\mathbb{B}_F$  is infinite, there is a sequence of literally disjoint  $F$ -positive subsets of  $\omega$ .*

*Proof.* Take the sequence  $(b_n)$  supplied by Lemma 8.3. Dependent choice implies countable choice, so select representatives  $C_n \subseteq \omega$  with  $q_F(C_n) = b_n$ . Define

$$B_n = C_n \setminus \bigcup_{j < n} C_j.$$

These sets are literally disjoint. Because  $b_n \wedge b_j = 0$  for  $j < n$ , their quotient classes satisfy

$$q_F(B_n) = b_n \wedge \bigwedge_{j < n} \neg b_j = b_n \neq 0.$$

The last inequality is exactly  $F$ -positivity by (15).  $\square$

**Theorem 8.5** (Local classification, ZF + DC). *Let  $F \in \mathcal{F}$  and let  $A$  have at least two elements. The following conditions are equivalent:*

- (i)  $\mathcal{R}_A$  is a sheaf on the full category of filters extending  $F$ , equipped with its dense topology;
- (ii) every matching family on every covering sieve at  $F$  has an amalgam;
- (iii) the Boolean algebra  $\mathbb{B}_F$  is finite.

The implication from (iii) to (i) is provable in ZF.

*Proof.* The first condition implies the second by definition, and Proposition 8.2 gives the third-to-first implication. If  $\mathbb{B}_F$  is infinite, Lemma 8.4 and Theorem 6.1 produce a matching family at  $F$  without an amalgam. This is the contrapositive of the remaining implication.  $\square$

For arbitrary  $A$ , the classification may be written

$$\mathcal{R}_A \text{ is a sheaf above } F \iff |A| \leq 1 \text{ or } \mathbb{B}_F \text{ is finite,} \quad (\text{ZF} + \text{DC}).$$

For  $F = \text{Fr}$ , the quotient is  $\mathcal{P}(\omega)/\text{Fin}$  and the explicit rows of Section 4 already give an infinite antichain, so dependent choice is unnecessary there.

The local theorem also identifies genuine positive cases. If a free ultrafilter  $U$  is given, the category above  $U$  has a single object;  $\mathbb{B}_U$  is the two-element Boolean algebra, and every  $\mathcal{R}_A$  is a sheaf on that one-object site. This does not conflict with failure on the larger category of all free filters.

## 9 What sheafification adds in classical set theory

### 9.1 The product of all ultrafilter values

This section assumes ZFC. We make the associated sheaf completely explicit, rather than relying on an abstract sheafification construction. For a free filter  $F$ , let

$$\text{Ult}(F) = \{U : U \text{ is an ultrafilter on } \omega \text{ and } F \subseteq U\}.$$

Define

$$\mathcal{S}_A(F) = \prod_{U \in \text{Ult}(F)} A^\omega / U. \quad (18)$$

When  $G \supseteq F$ , its ultrafilter extensions form a subset of  $\text{Ult}(F)$ , and restriction in  $\mathcal{S}_A$  is the corresponding projection. There is a natural transformation

$$\eta_F : \mathcal{R}_A(F) \longrightarrow \mathcal{S}_A(F), \quad [f]_F \longmapsto ([f]_U)_{U \in \text{Ult}(F)}. \quad (19)$$

**Lemma 9.1.** *The ultrafilters extending  $F$  form a covering sieve at  $F$ . Every covering sieve at  $F$  contains every member of  $\text{Ult}(F)$ . Each map  $\eta_F$  is injective.*

*Proof.* Every proper filter extends to an ultrafilter by the ultrafilter lemma, which is available in ZFC. Therefore these maximal extensions are dense. A proper extension of an ultrafilter equals itself, so this dense family is extension-closed and hence a sieve.

If  $D$  is any covering sieve and  $U \in \text{Ult}(F)$ , density applied to  $U$  gives a member of  $D$  extending  $U$ . It must be  $U$  itself.

For injectivity, suppose  $[f]_F \neq [g]_F$ . Then  $E(f, g) \notin F$ , so  $\langle F \cup \{E(f, g)^c\} \rangle$  is proper. Extend it to an ultrafilter  $U$ . Its containing  $E(f, g)^c$  prevents  $E(f, g) \in U$ , and the  $U$ -coordinates of the two images differ.  $\square$

**Theorem 9.2** (Explicit associated sheaf, ZFC). *The presheaf  $\mathcal{S}_A$  is a sheaf, and  $\eta : \mathcal{R}_A \rightarrow \mathcal{S}_A$  has the universal property of sheafification.*

*Proof.* First let  $(s_G)_{G \in D}$  be a matching family for  $\mathcal{S}_A$  on a cover of  $F$ . For each  $U \in \text{Ult}(F)$ , Lemma 9.1 gives  $U \in D$ . Since  $U$  has only itself as an ultrafilter extension,  $\mathcal{S}_A(U) = A^\omega/U$ . Let the  $U$ -coordinate of  $s \in \mathcal{S}_A(F)$  be  $s_U$ . For  $G \in D$  and  $U \supseteq G$ , compatibility says that the  $U$ -coordinate of  $s_G$  is  $s_U$ . Hence  $s$  restricts to  $s_G$ . The same coordinates show uniqueness. This proves sheafness.

Now let  $\mathcal{T}$  be any sheaf and  $\varphi : \mathcal{R}_A \rightarrow \mathcal{T}$  any morphism of presheaves. Given  $s \in \mathcal{S}_A(F)$ , assign the section

$$\varphi_U(s(U)) \in \mathcal{T}(U) \quad \text{to each } U \in \text{Ult}(F).$$

Distinct ultrafilters are incompatible, so these sections form a matching family on the ultrafilter cover. Glue them uniquely in  $\mathcal{T}(F)$  and call the result  $\psi_F(s)$ . These definitions commute with restriction: both possible restrictions at a stronger filter have the same values at all its ultrafilter extensions and are therefore equal by sheaf uniqueness. Thus  $\psi$  is a morphism.

For  $s = \eta_F([f]_F)$ , naturality of  $\varphi$  shows that the sections being glued are the restrictions of  $\varphi_F([f]_F)$ . Hence  $\psi\eta = \varphi$ . Conversely, any morphism with this property must take  $s$  to a section with these prescribed ultrafilter restrictions; sheaf uniqueness forces that section to be  $\psi_F(s)$ . This proves the universal property.  $\square$

This proof also explains why sheafification does not collapse unequal old sections: every  $\eta_F$  is injective. It supplies additional sections by allowing independent values at maximal filters, not by identifying distinct equivalence classes already present.

## 9.2 The two-element case: clopen sets versus arbitrary sets

Let  $A = 2 = \{0, 1\}$ . Every sequence  $f : \omega \rightarrow 2$  is a characteristic function. At an ultrafilter  $U$ , exactly one of its two fibres lies in  $U$ , so  $2^\omega/U$  is canonically 2. Accordingly,

$$\mathcal{S}_2(F) = 2^{\text{Ult}(F)} \cong \mathcal{P}(\text{Ult}(F)), \quad \mathcal{R}_2(F) \cong \mathbb{B}_F. \quad (20)$$

The second identification sends a characteristic function to the quotient class of its support; equality of functions modulo  $F$  means that the symmetric difference of their supports belongs to  $I_F$ .

For  $X \subseteq \omega$ , set

$$\widehat{X} = \{U \in \text{Ult}(F) : X \in U\}.$$

These sets give the usual Stone topology on  $\text{Ult}(F)$ : each is both open and closed, finite intersections and complements have the expected Boolean formulas, and they form a base. For completeness, compactness follows directly from the ultrafilter lemma. A family of these basic sets with no finite subcover has complements whose representatives, together with  $F$ , have the finite-intersection property; an ultrafilter

extending that family is missed by the alleged cover. A general open cover can be refined by the basic sets. Thus this space is compact. Every clopen subset, being compact and open, is a finite union of basic sets and is therefore itself  $\widehat{X}$  for some  $X$ .

Under (20),  $\eta_F$  is therefore the inclusion

$$\mathbb{B}_F \cong \text{Clop}(\text{Ult}(F)) \hookrightarrow \mathcal{P}(\text{Ult}(F)). \quad (21)$$

This is a description of the sheafification map, not a claim that the power set on the right is the Boolean completion in the order-dense or MacNeille sense. Those are different requirements.

For any finite nonempty  $A$ , the same description reads: the old sections induce the continuous maps from the Stone space to discrete  $A$ , while the associated sheaf permits all maps. Each fibre of the map induced by a sequence is a basic clopen set. Conversely, a continuous map gives a finite clopen partition; representing its pieces in  $\mathcal{P}(\omega)$  and removing the finitely many negligible overlaps produces a sequence with precisely those ultrafilter values.

### 9.3 The missing section in the explicit example

Return to  $F = \text{Fr}$  and the rows and tails of Section 3. Define

$$K = \{U \in \text{Ult}(\text{Fr}) : F_\infty \subseteq U\} = \bigcap_m \widehat{T_m}. \quad (22)$$

Every free ultrafilter contains at most one row. If it contains no row, it contains the complement of each row and therefore all  $T_m$ . Hence every free ultrafilter either extends a unique  $F_n$  or belongs to  $K$ . The two-valued matching family becomes the function

$$e(U) = \begin{cases} 0, & U \supseteq F_n \text{ for some } n, \\ 1, & U \in K. \end{cases}$$

It is the characteristic function of  $K$  in  $\mathcal{S}_2(\text{Fr})$ .

**Proposition 9.3.** *The set  $K$  is closed but not clopen. Equivalently, the section  $e$  is not in the image of  $\eta_{\text{Fr}}$  for  $A = 2$ .*

*Proof.* Equation (22) makes  $K$  closed. Suppose  $K = \widehat{E}$  for some  $E \subseteq \omega$ . Then  $E$  lies in every ultrafilter extending  $F_\infty$ , which forces  $E \in F_\infty$ : otherwise one could extend  $\langle F_\infty \cup \{E^c\} \rangle$  to an ultrafilter. Thus  $T_m \setminus E$  is finite for some  $m$ .

Take an ultrafilter  $U \supseteq F_m$ , which exists in ZFC. Since  $B_m \subseteq T_m$  and  $B_m \setminus E$  is finite, we have  $E \in F_m \subseteq U$ . Hence  $U \in \widehat{E} = K$ . But no proper filter extends both  $F_m$  and  $F_\infty$ , a contradiction. Thus  $K$  is not clopen.  $\square$

Closed but nonclopen does not imply that  $K$  has empty interior. Indeed, one can choose the explicit selector  $S = \{2^n - 1 : n < \omega\}$ , which meets each row once and is almost contained in every  $T_m$ . The nonempty basic open set  $\widehat{S}$  is contained in  $K$ . This observation helps avoid a misleading picture of the tail component as necessarily nowhere dense.

## 10 Consequences for the hyperreal setting

When  $A = \mathbb{R}$ , pointwise ring operations make the reduced powers and their restriction maps into rings and ring homomorphisms. At an ultrafilter  $U$  the reduced power is the usual ultrapower; over the whole filter site it is a varying family, not one fixed field. The underlying set obstruction uses only  $0, 1 \in \mathbb{R}$ , so no ordered-field property, transfer theorem, saturation theorem, or cardinal estimate for  $\mathbb{R}$  is required.

In the product sheaf of Section 9, the same coordinatewise 0–1 element  $e$  satisfies  $e^2 = e$ . It does not come from any sequence of reals. To see this directly, if  $\eta([f]) = e$ , then at every row filter the injectivity in Lemma 9.1 forces  $[f]_{F_n} = [0]_{F_n}$ , and at the tail filter it forces  $[f]_{F_\infty} = [1]_{F_\infty}$ . Proposition 3.3 rules this out.

This is a *new global idempotent section relative to the original presheaf*, not a claim that its old rings had no nontrivial idempotents. For example, characteristic sequences already give idempotents in  $\mathbb{R}^\omega/\text{Fr}$ . Nor is the new global section a nontrivial idempotent inside one hyperreal field: at each ultrafilter it is just 0 or 1. The product of many fields is not itself a field.

Three statements should therefore be kept separate. Local equality of old sections detects equality at the base filter; this is separatedness. Compatible local sections need not have a single representative; this is the obstruction. Passing to the associated sheaf supplies such sections but changes the available global elements. None of these statements contradicts the possibility-semantic account in the source, which already distinguishes its presheaf construction from a sheaf model [1, Section 5.2].

The result also depends on the category and its topology. It is not an assertion that every sheaf-based construction of nonstandard analysis fails, or that every reduced-power presheaf on a different filter site has the same behavior. The relative countably generated site considered in Section 7 was analyzed separately; no unexamined change of topology is implicit in the main theorem.

## 11 Reproducibility and proof audit

### 11.1 Exact supplementary checks

The mathematical proof is infinite and does not rest on numerical experiments. The archive nevertheless supplies executable checks of the arithmetic identities and of a finite analogue of the pasting theorem. All code uses the Python standard library and exact integers. Running

```
python3 verify.py --output verification_results.json
```

reproduces the following results:

| Check                                | Number checked |
|--------------------------------------|----------------|
| Row–tail membership identities       | 150,000        |
| Exact cutoff-to-witness certificates | 64,925         |
| Invalid inputs correctly rejected    | 6              |
| Finite generator families            | 717            |
| Finite local configurations          | 86,669         |
| Compatible finite configurations     | 9,209          |
| Unique gluings for finite covers     | 7,421          |
| Compatible noncover pastings         | 1,788          |

All checks passed. Of the cutoff certificates, 54,925 exhaust  $m = 0, \dots, 12$  and  $N_0, N_1 = 0, \dots, 64$ ; another 10,000 use a fixed pseudorandom seed,  $m \leq 200$ , and 512-bit nonnegative cutoffs. The program also checks that the returned witness is the least possible one.

For the finite-pasting tests, a nonempty subset  $K$  of a finite index set  $I$  determines the principal filter  $F_K = \{X \subseteq I : K \subseteq X\}$ . A reduced-power class is exactly a function on  $K$ . The program exhausts families of at most three distinct nonempty kernels, all local functions, and all global functions for  $|I| \leq 4$ ,  $|A| = 2$ , and  $|I| \leq 3$ ,  $|A| = 3$ . Compatibility is checked on pairwise intersections. The least-index pasting construction is then compared with an independent enumeration of global solutions. A family covers the weakest proper filter exactly when the union of its kernels is  $I$ ; in that case uniqueness is checked.

A finite index set has no proper filter containing all of its cofinite subsets, since every subset, including the empty set, is cofinite there. These finite tests are therefore *not* finite objects of the Fréchet site. They test the more general finite-pasting theorem of Section 5, which was deliberately stated for arbitrary proper filters on an index set. The tests cannot establish sheaf failure on an infinite site, and no proof-assistant verification is claimed.

## 11.2 Foundational dependencies

| Result                                                 | Assumptions used |
|--------------------------------------------------------|------------------|
| Explicit counterexample and global classification      | ZF               |
| Separatedness and finite-pasting theorem               | ZF               |
| Positive-row obstruction at a fixed base               | ZF               |
| Failure at every countably generated free base         | ZF               |
| Finite Boolean quotient implies local sheafness        | ZF               |
| Infinite Boolean quotient gives positive rows          | ZF + DC          |
| Product-ultrapower associated sheaf; Stone description | ZFC              |

The stronger assumptions in the final rows are not smuggled back into the first rows. In particular, the main construction does not choose any ultrafilter. The ZFC argument is an interpretation and extension of a result already established in ZF.

## 11.3 The main argument as a checklist

The decisive checks are all explicit. The row and tail families are proper filters; their incompatibility follows from displayed disjoint members. The cover includes *all* extensions of each generator. For every test filter, a positive row can be adjoined, or all finite row tails already belong to it. Each local section is the class of a specified constant sequence. Incompatibility makes the assignment unambiguous and persistent. Finally, membership at the tail filter produces one tail index, and membership at that row filter produces the contradiction on an infinite set.

No step requires that a matching family be globally represented before trying to glue it. No countable intersection is asserted to belong to a filter. The empty-domain convention is handled explicitly. These are the main points at which an apparently similar argument could fail; they are also the points against which the proof was checked.

## 12 Conclusion and the remaining status questions

The selected published question has the following answer within the argument of this manuscript: uncountability is not necessary. Two values suffice, and a value set has a sheaf reduced-power presheaf on the full Fréchet site exactly when it has at most one element. The proof uses an explicit row partition and a tail filter and is available in ZF.

The extensions identify the mechanism rather than merely producing an isolated counterexample. Finite systems always paste. Countably generated free bases always admit the obstruction. With dependent choice, the finite or infinite nature of the quotient Boolean algebra completely determines local sheafness for a nontrivial value set. In classical set theory, the associated sheaf replaces the requirement of a single sequence by a compatible product of ultrapower values, and the missing section is already binary.

The mathematical arguments are presented as complete proofs, but this manuscript has not undergone independent expert review or formal verification. The literature check establishes a documented open question in 2023 and its repetition in 2024; it does not establish that no later or unpublished proof exists. Those limitations concern certification and priority, not an omitted step attributed to the problem's age or difficulty.

A precise further foundational question suggested by the proofs is whether the infinite-Boolean-quotient direction of Theorem 8.5 can be established in ZF without additional hypotheses on the base filter. This manuscript does not assert that the use of dependent choice there is optimal, nor that this suggested question is new or open in the literature. It records exactly where choice enters the present extension.

### A A compact standalone proof

Let  $\omega = \bigsqcup_n B_n$  be a partition into infinite sets and  $T_m = \bigcup_{n \geq m} B_n$ . Let  $F_n$  consist of sets containing all but finitely many elements of  $B_n$ , and let  $F_\infty$  consist of sets containing all but finitely many elements of some  $T_m$ . All are proper free filters. Distinct  $F_n$  are incompatible, and  $F_n$  is incompatible with  $F_\infty$  because  $B_n \cap T_{n+1} = \emptyset$ .

The sieve of all extensions of these filters covers Fr. Indeed, if a free filter  $G$  has a positive row, adjoining it gives an extension in the sieve. Otherwise  $G$  contains every row complement, hence every  $T_m$ , so  $G$  itself extends  $F_\infty$ .

For distinct  $a_0, a_1 \in A$ , assign the constant  $a_0$  class on row components and the constant  $a_1$  class on the tail component. Incompatibility makes this a matching family. A gluing sequence  $f$  would equal  $a_1$  cofinitely on some  $T_m$  and equal  $a_0$  cofinitely on  $B_m$ . Since  $B_m \subseteq T_m$  is infinite, this is impossible. For  $|A| \leq 1$ , the sheaf condition is immediate. This proves the classification without any choice axiom.

### B Source map and archive contents

The exact source locations are as follows. In [1], the filter category, covers, and reduced-power presheaf are Definitions 5.4, 5.5, and 5.7 on printed page 515. The sufficient cardinality condition is Fact 5.10 on pages 516–517. The motivating question is the paragraph immediately after its proof on page 517. The 2024 version is [2, Section 7.4.2, Fact 7.4.10 and following paragraph], with the question on printed page 249. The PDF page indices are respectively 27 and 259 when counted from zero. These passages were checked against page images as well as extracted text.

The archive contains `article.tex`, `article.pdf`, the bibliographic database `references.bib`, the

two Python programs, their actual `verification_results.json`, a `README.md`, a separate proof audit and literature-search record, and a build script. Source publications and font files are not bundled. The PDF bibliography is self-contained; the BibTeX database is supplied for reuse and is not required by the build.

## References

- [1] Guillaume Massas. A Semi-Constructive Approach to the Hyperreal Line. *Australasian Journal of Logic* **20**(3) (2023), 490–536, article 6. DOI: 10.26686/ajl.v20i3.8254. <https://ojs.victoria.ac.nz/ajl/article/view/8254>. Open question: p. 517, after Fact 5.10.
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